

Regression Analysis

You have a large enough capital to build a factory:

- What kind of product?
- Is there enough demand?
- Market research?
 - Can we explain the variability of demand with some other factors?
 - Can we predict the future demand to justify the construction of a new plant?

Translation

- **regression analysis:** *bağlanım analizi. İstatistikte belirli rasgele değişkenlerin başka değişkenlere olan bağıntısının irdelenmesi ve bağımlı değişkenlerin bağımsızlardan kestirimi analizi. (Fra. analyse de régression)*
- **regression curve:** *bağlanım eğrisi. Bir rasgele değişkenin beklentisini bir ya da daha çok başka değişkene doğrusal-olmayan biçimde ilişkilendiren denklem. Bağlanım eşitliğinin grafik gösterimi üç boyutlu olduğunda bağlanım yüzeyi adını alır. (Fra. courbe de régression)*
- **regression equation:** *bağlanım denklemi. Bir rasgele değişkenin beklentisini bir ya da daha çok başka değişkene ilişkilendiren denklem. (Fra. relation régressive)*
- **regression line:** *bağlanım doğrusu. Yalnızca doğrusal bağlanımı gösteren bağlanım eğrisi. (Fra. ligne de régression)*
- **regression:** *bağlanım. Bir değişkeni öteki değişkenlere bağımlı kılma işi ve biçimi. (Fra. régression)*

Simple linear regression (1)

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

independent
variable

dependent
variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

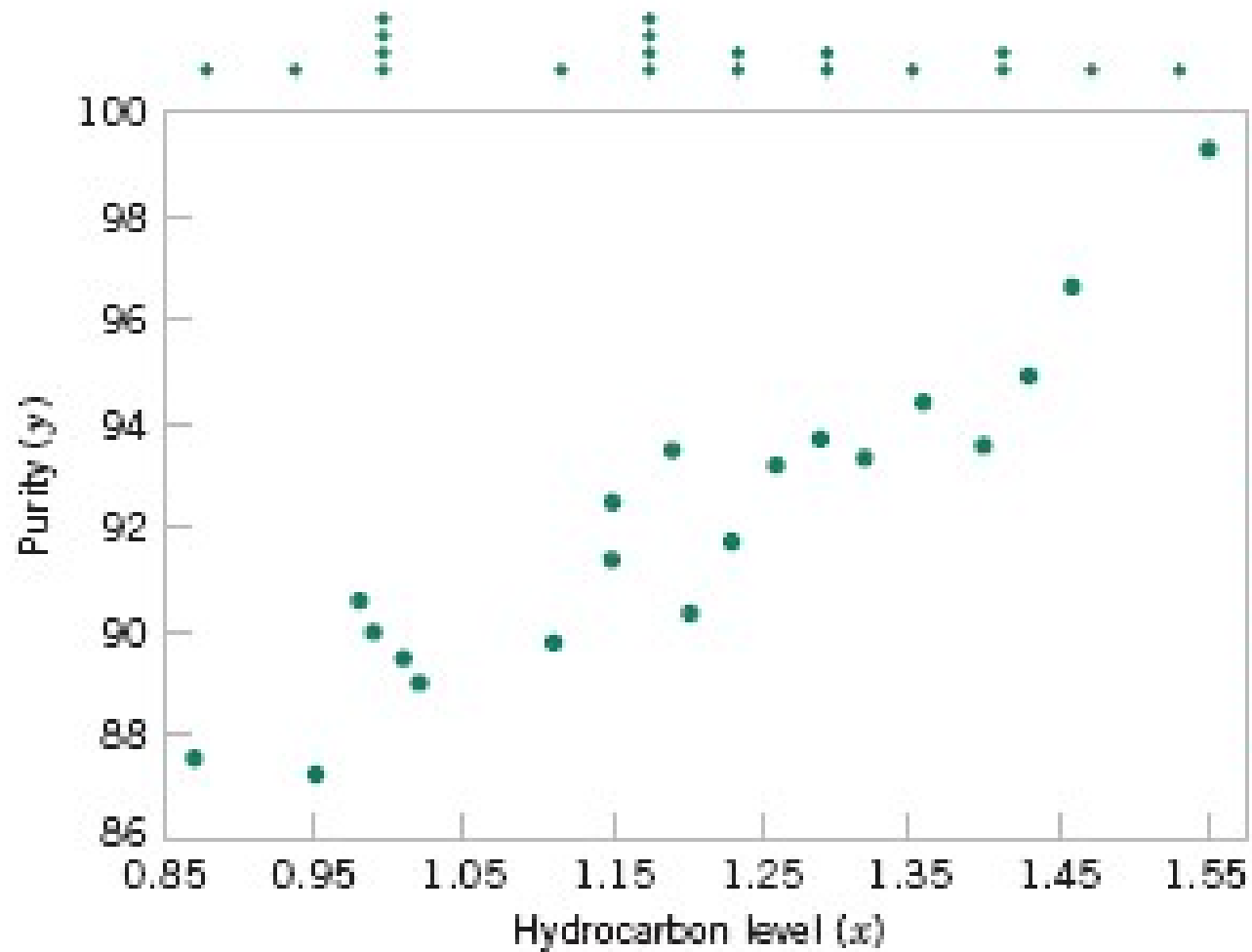
$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

Empirical Models

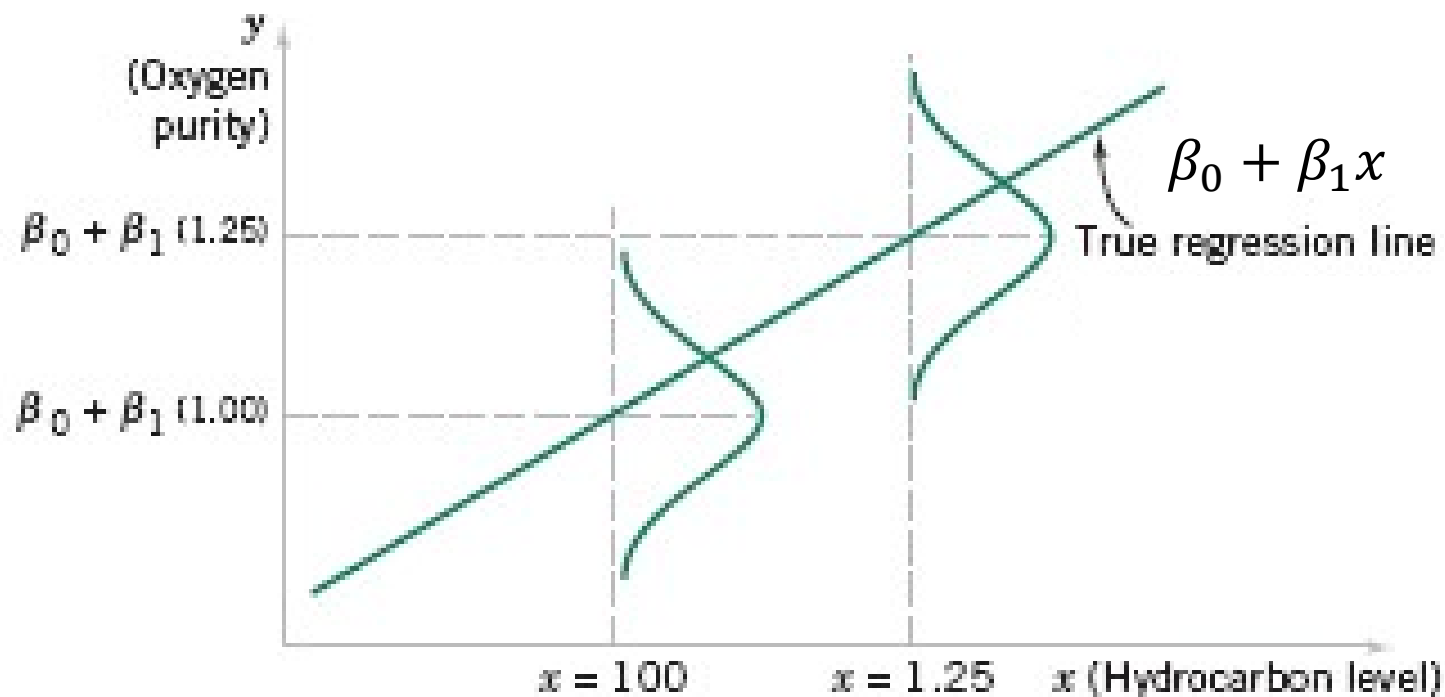
Observation Number	Hydrocarbon Level $x(\%)$	Purity $y(\%)$
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

Empirical Models



Scatter Diagram of oxygen purity versus hydrocarbon level.

Empirical Models



The distribution of Y for a given value of x for the oxygen purity-hydrocarbon data.

Simple linear regression (2)

$$E(\beta_0 + \beta_1 x^* + \varepsilon) = \underbrace{\beta_0 + \beta_1 x^*}_{\substack{\text{Expected change} \\ \text{in } Y \text{ with a 1-unit} \\ \text{increase in } x.}}$$

The mean value of Y is a linear function of x.

$$V(\beta_0 + \beta_1 x^* + \varepsilon) = \sigma^2$$

$$Y \sim N(\beta_0 + \beta_1 x^*, \sigma) \quad \text{at } x^*$$

$$\varepsilon \sim N(0, \sigma^2)$$

Estimating model parameters: β_0 & β_1 (1)

$$\begin{array}{l} b_0 = \hat{\beta}_0 \\ b_1 = \hat{\beta}_1 \end{array} \quad \Longrightarrow \quad \begin{array}{l} \hat{y} = b_0 + b_1 x \\ y_i = \overbrace{b_0 + b_1 x_i}^{\hat{y}_i} + \underbrace{e_i}_{\text{residual}} \end{array}$$

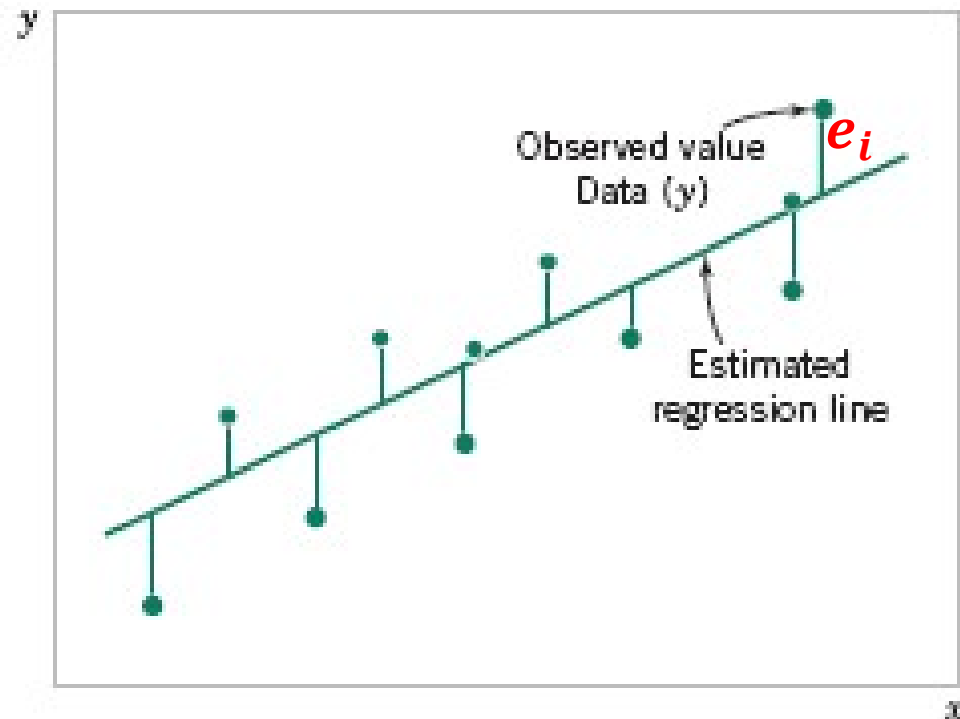
$\hat{y}_i$: estimate for the i^{th} observation (x_i, y_i) - fitted or predicted values for x_i

want to minimize the “errors” ...

Simple Linear Regression

- The method of least squares is used to estimate the parameters, β_0 and β_1 by minimizing the sum of the squares of the vertical deviations in the figure.

Deviations of the data from the estimated regression model.



Minimizing the squared errors in simple linear regression

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$$

$$\left. \frac{\partial SSE}{\partial b_0} \right|_{b_0, b_1} = -2 \sum_{i=1}^n (y_i - (b_0 + b_1 x_i)) = 0$$

$$\left. \frac{\partial SSE}{\partial b_1} \right|_{b_0, b_1} = -2 \sum_{i=1}^n (y_i - (b_0 + b_1 x_i)) x_i = 0$$

Least square estimates

$$b_1 = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2} = \frac{S_{xy}}{S_{xx}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$\bar{y} = \frac{\sum y_i}{n}, \bar{x} = \frac{\sum x_i}{n}$$

$$S_{xy} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n y_i(x_i - \bar{x}) = \sum x_i y_i - \frac{\sum x_i \sum y_i}{n}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i(x_i - \bar{x}) = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

Fitted or estimated regression line

The **fitted** or **estimated regression line** is therefore

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Note that each pair of observations satisfies the relationship

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i, \quad i = 1, 2, \dots, n$$

where $e_i = y_i - \hat{y}_i$ is called the **residual**. The residual describes the error in the fit of the model to the i th observation y_i . Later in this chapter we will use the residuals to provide information about the adequacy of the fitted model.

Another way to look at b_1

$$b_1 = \frac{S_{xy}}{S_{xx}} = r \frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

sample correlation coefficient: $r_{xy} = r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$

$$S_{xy} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})$$

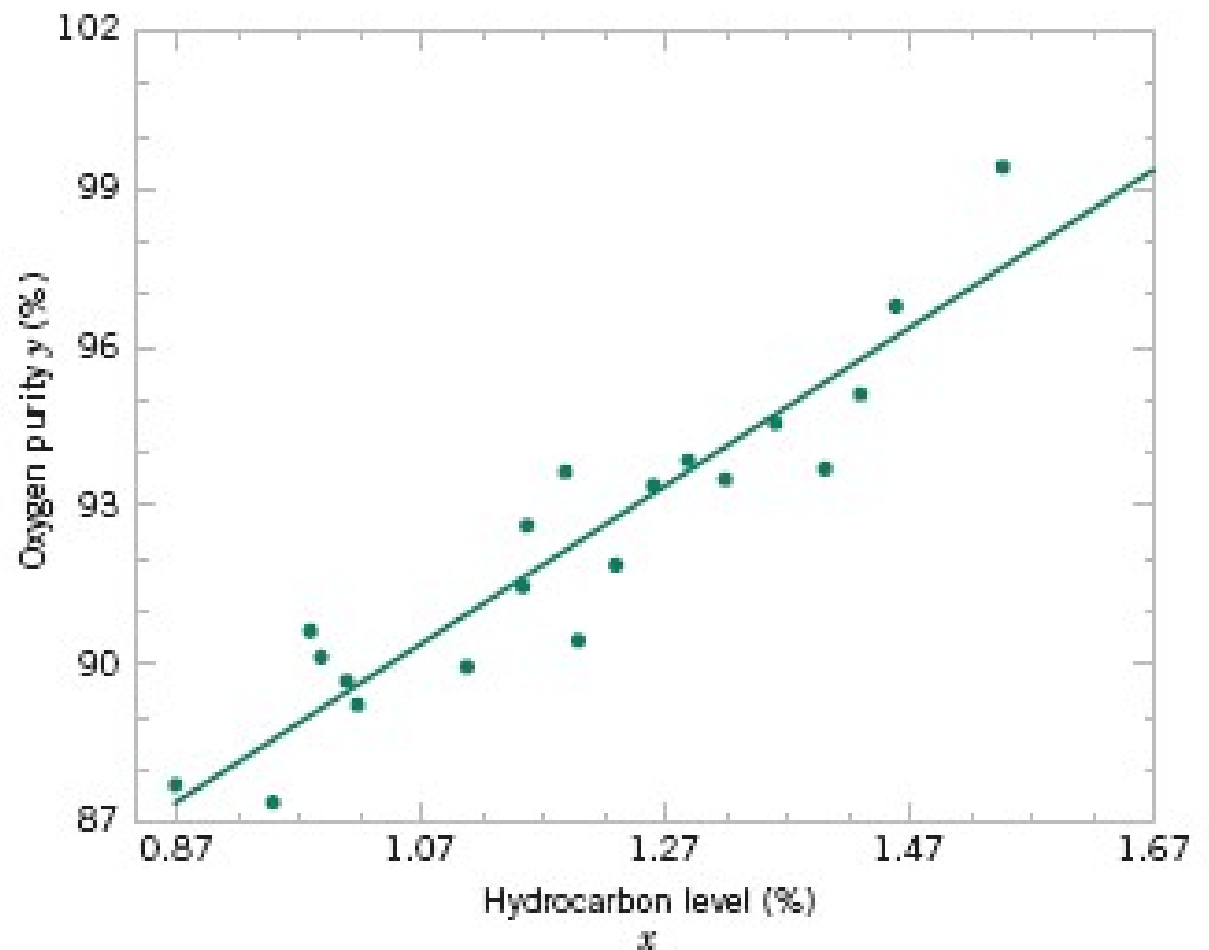
$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$$

Example: Oxygen Purity

What does this equation mean?

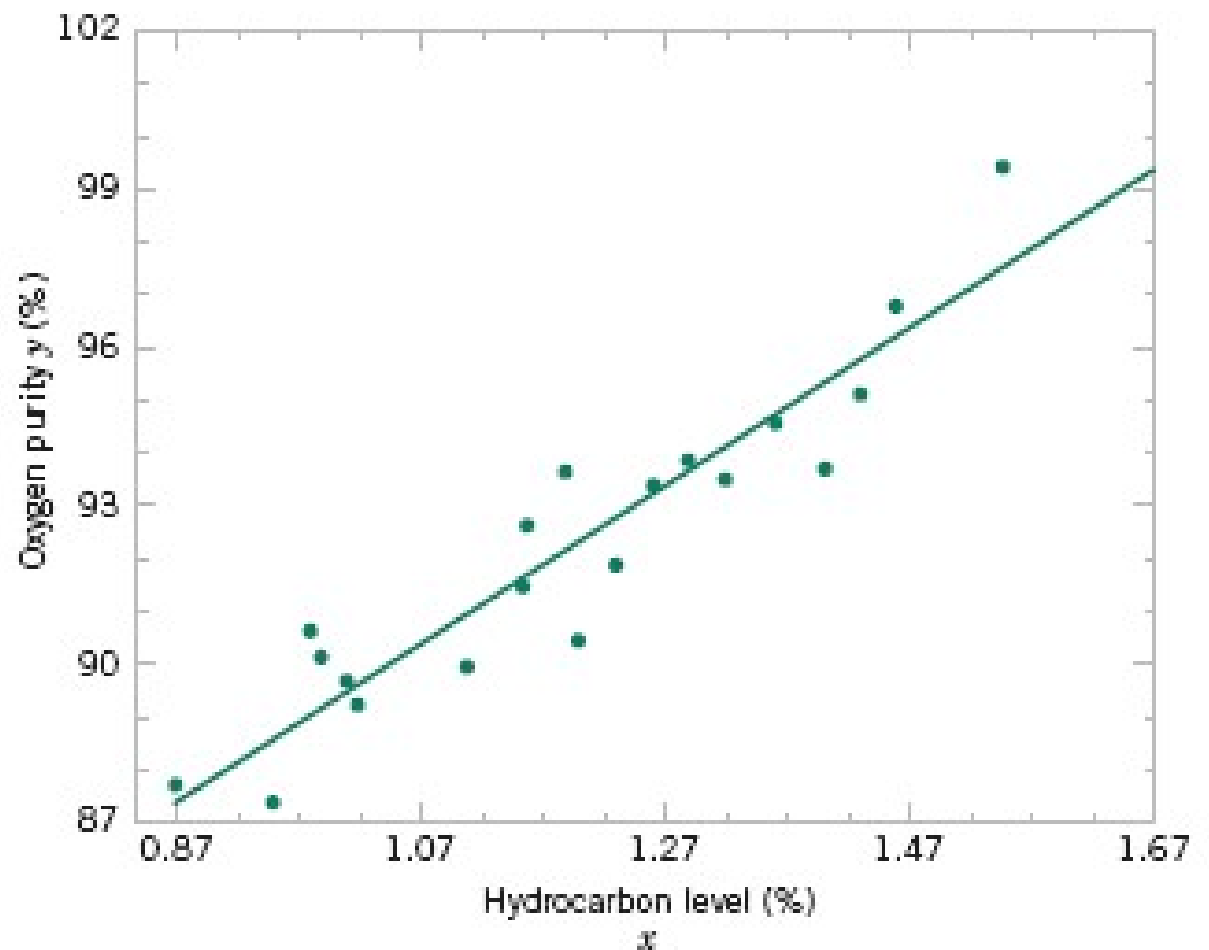
Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.20 + 15x$.



Example: Oxygen Purity

If hydrocarbon level (%) is 1, ?

Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.20 + 15x$.

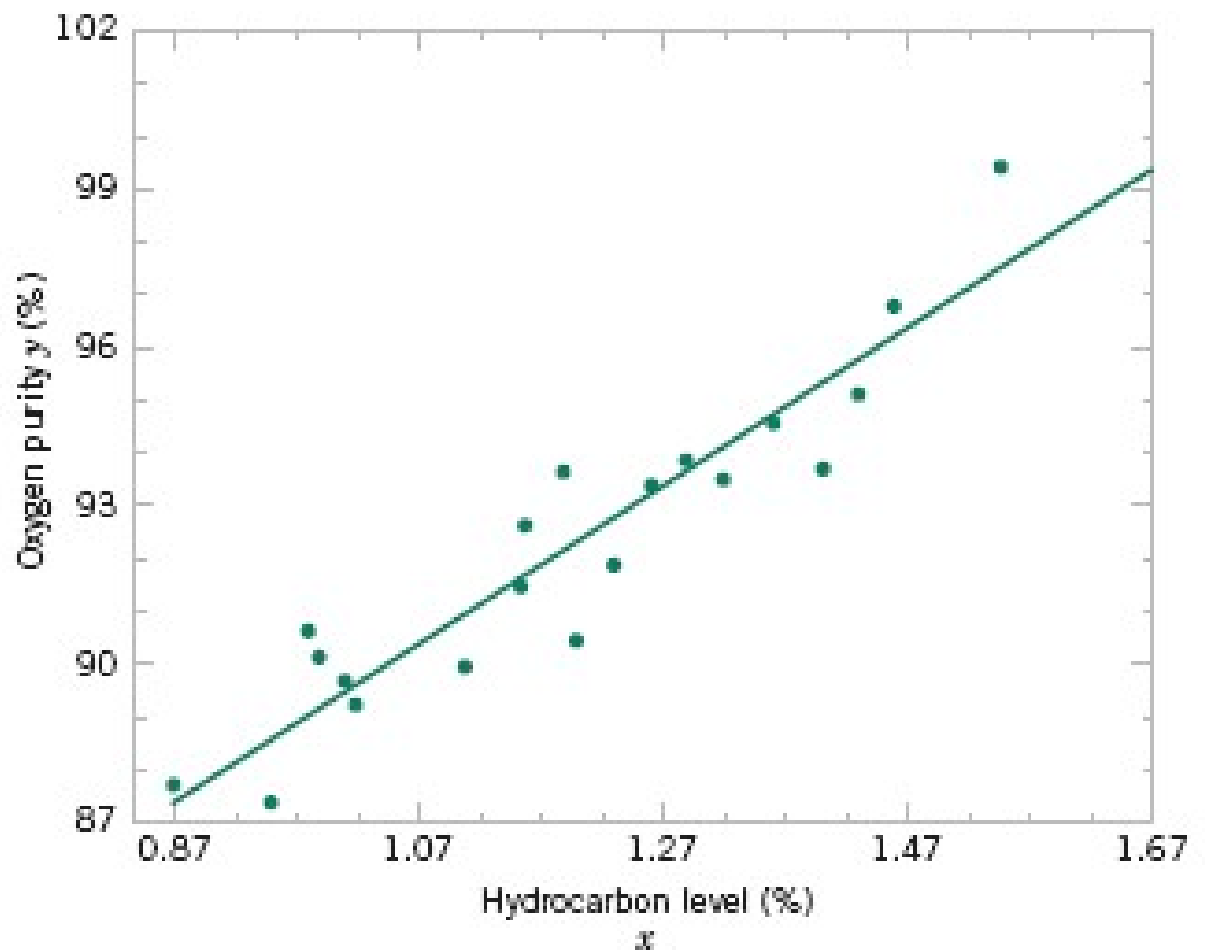


Example: Oxygen Purity

extrapolation:
dangerous!!!

If hydrocarbon level (%) is 0.2 or 3, ?

Scatter plot of oxygen
purity y versus
hydrocarbon level x
and regression model
 $\hat{y} = 74.20 + 15x$.



Example: Oxygen Purity

Regression Analysis

The regression equation is

$$\text{Purity} = 74.3 + 14.9 \text{ HC Level}$$

Predictor	Coef	SE Coef	T	P
Constant	74.283 $\leftarrow \hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 $\leftarrow \hat{\beta}_1$	1.317	11.35	0.000

S = 1.087

R-Sq = 87.7%

R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 $\leftarrow SS_E$	1.18 $\leftarrow \hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

Values of Predictors for New Observations

New Obs	HC Level
1	1.00

Example: Python Output

	coef	std err	t	P> t	[0.025	0.975]
	$b_0 = \hat{\beta}_0$					
Intercept	74.283	1.593	46.62	0.000	71.083	77.603
X	14.947	1.317	11.35	0.000	12.346	17.448
	$b_1 = \hat{\beta}_1$					

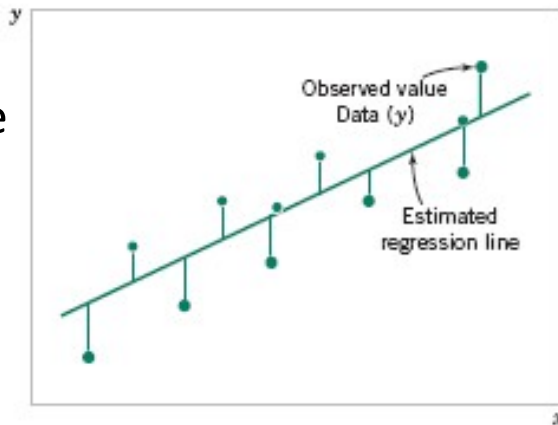
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

What are we interested in?

Building the model

- Estimating β_0 and β_1
→ the fitted regression line

b_0, b_1 : Minimize the sum of squared deviations.



Evaluating the model

- How useful is the model?
- Is the model significant?

- Estimating σ
- Distribution of $b_0, b_1, \hat{Y} = b_0 + b_1 x$

How much of the variability in Y is explained by X?

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

- LR model: $Y = \underbrace{\beta_0 + \beta_1 X}_{\text{True regression line}} + \overset{\text{Random errors}}{\varepsilon} \quad \varepsilon \sim N(0, \sigma^2)$

$$y_i = \underbrace{b_0 + b_1 x_i}_{\substack{\text{Estimated} \\ \text{regression line}}} + \underset{\text{residuals}}{e_i}$$

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$$

ANalysis Of VAriance identity

$$SST = S_{yy} = \sum (y_i - \bar{y})^2$$

total variability of Y.

$$SSE = \sum (y_i - \hat{y}_i)^2$$

error sum of squares

Interpretation: unexpected variability of Y in regression

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

regression sum of squares

Interpretation: amount of variability that is explained by the model

Goodness-of-fit
statistics

$$SST = \sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$

ANalysis Of VAriance identity

Goodness-of-fit
statistics

$$SST = \sum (y_i - \bar{y})^2 = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

$$\frac{\sum (y_i - \bar{y})^2}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{SST} + \frac{\sum (y_i - \hat{y}_i)^2}{SST}$$

$$1 = r^2 + \frac{SSE}{SST}$$

How much of the variability in Y is explained by X?

- The sample: $\{(x_i, y_i)\}_{i=1}^n$
- LR model: $Y = \beta_0 + \beta_1 X + \varepsilon, \varepsilon \sim N(0, \sigma^2)$

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST} \longrightarrow$$

Proportion of observed y variation that can be explained by the simple linear regression model.
measures the degree of **linear** relationship between X and Y

Coefficient of determination

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$



Proportion of observed y variation that can be explained by the simple linear regression model.
measures the degree of **linear** relationship between X and Y

r^2 has no relation with the magnitude of the slope of regression line.

Interpretation of r

- r^2 is not related with the magnitude of the slope.
- When r^2 is small, an alternative model should be sought: Generally, one looks for other independent variables which can explain a bit more of the variation in Y .

→ multiple regression

- Interpretation of value of r^2 changes with the context:
 - physical/engineering: 80%
 - social/psychology: 50% can be good enough

Linear regression model

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

- LR model:
$$Y = \underbrace{\beta_0 + \beta_1 X}_{\text{True regression line}} + \overset{\text{Random errors}}{\varepsilon} \quad \varepsilon \sim N(0, \sigma^2)$$

$$y_i = \underbrace{b_0 + b_1 x_i}_{\substack{\text{Estimated} \\ \text{regression line}}} + \underset{\text{residuals}}{e_i}$$

Distribution of b_0 ?

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$E(b_0) = \beta_0$$

$$V(b_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

$$b_0 \sim N \left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)$$

But σ^2 is not known...

Distribution of b_1 ?

$$b_1 = \frac{S_{xy}}{S_{xx}}$$

$$E(b_1) = \beta_1$$

$$V(b_1) = \frac{\sigma^2}{S_{xx}}$$

$$\Rightarrow b_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

But σ^2 is not known...

Estimating σ^2 (1)

$$\varepsilon_i \sim N(0, \sigma^2)$$

variability of each of Y_i about the true regression line.

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

variability of data pts about the fitted regression line.

$$\hat{\sigma}^2 = ?$$

Estimating σ^2 (2)

*two parameters to
estimate β_0, β_1*

$$E[SSE] = (n - 2)\sigma^2 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{SSE}{n - 2} = MSE$$

can be shown

Two parameters to
Estimate β_0, β_1 $\Rightarrow$ a loss of 2 df

MSE is an unbiased estimator for σ^2 .

Example: Oxygen Purity

Regression Analysis

The regression equation is

$$\text{Purity} = 74.3 + 14.9 \text{ HC Level}$$

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R-Sq = 87.7%

R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
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Residual Error	18	21.25 $\leftarrow SS_E$	1.18 $\leftarrow \hat{\sigma}^2$		
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New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

Values of Predictors for New Observations

New Obs	HC Level
1	1.00

Example: Python Output

	df	sum_sq	mean_sq	F	PR(>F)
X	1.0	152.13	152.13	128.86	0.000
Residual	18.0	21.25	1.18	NaN	NaN
		SSE	MSE		

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).

Regression Analysis: Rating versus Meters

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	1669.02	1669.02	61.02	0.000
Error	30	820.50	27.35		
Total	31	2489.52			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
5.22972	67.04%	65.94%	62.11%

Coefficients

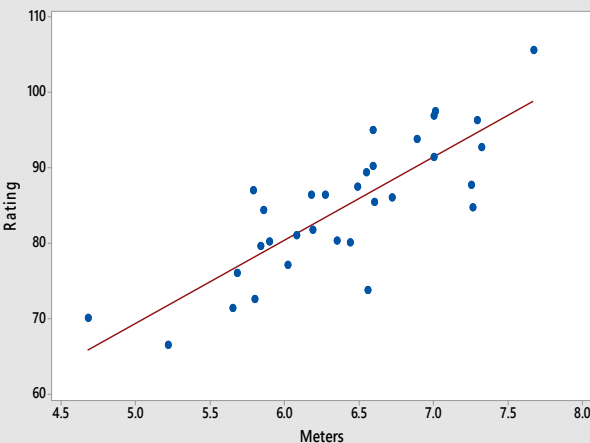
Term	Coef	SE Coef	T-Value	P-Value
Constant	14.19	9.09	1.56	0.129
Meters	11.04	1.41	7.81	0.000

Regression Equation

$$\text{Rating} = 14.19 + 11.04 \text{ Meters}$$

What is the estimate of σ^2 ?

Scatterplot of Rating vs Meters



Inferences about slope, β_1 (1)

- If $\beta_1 = 0$, what is the meaning of regression?
- $b_1 = \hat{\beta}_1 \Rightarrow$ inferences for β_1 through analysis of b_1 .
- **Distribution of b_1 :**

$$b_1 = \frac{S_{xy}}{S_{xx}} \Rightarrow b_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

But σ^2/S_{xx} is not known...

Inferences about slope, β_1 (2)

σ^2 is not known, but can be estimated with $df=n-2$:

$$\hat{V}(b_1) = \hat{\sigma}_{b_1}^2 = s_{b_1}^2 = \frac{\hat{\sigma}^2}{S_{xx}} \quad \Rightarrow \quad s_{b_1} = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

$$\Rightarrow \quad T = \frac{b_1 - \beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} = \frac{b_1 - \beta_1}{s_{b_1}} \sim t_{n-2}$$

Confidence interval for β_1

- Straightforward when the distribution of β_1 is known as t_{n-2} ...
- Your book has the details ...

Hypothesis testing on β_1

$$H_0: \beta_1 = \beta_1^0$$

$$\text{Test statistics: } T = \frac{b_1 - \beta_1^0}{s_{b_1}} \sim t_{n-2}$$

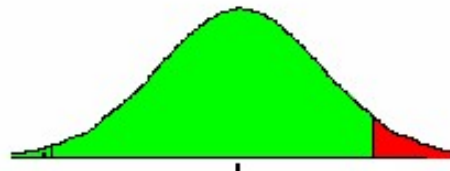
$$H_a: \beta_1 > \beta_1^0$$

$$H_a: \beta_1 \neq \beta_1^0$$

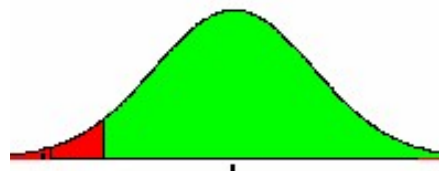
$$H_a: \beta_1 < \beta_1^0$$

 Accept

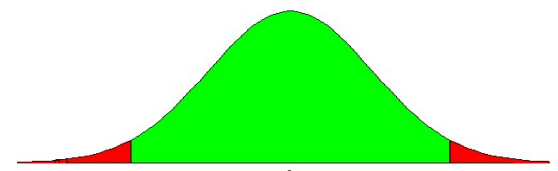
 Reject



$$\text{R.R.: } t \geq t_\alpha$$



$$\text{R.R.: } t \leq -t_\alpha$$



$$\begin{aligned} \text{R.R.: } t &\leq -t_{\alpha/2} \\ \text{or } t &\geq t_{\alpha/2} \end{aligned}$$

Significance of regression analysis

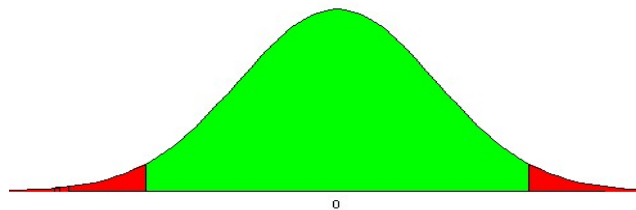
$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

$$\text{Test statistics: } T = \frac{b_1}{S_{b_1}} \sim t_{n-2}$$

 Accept

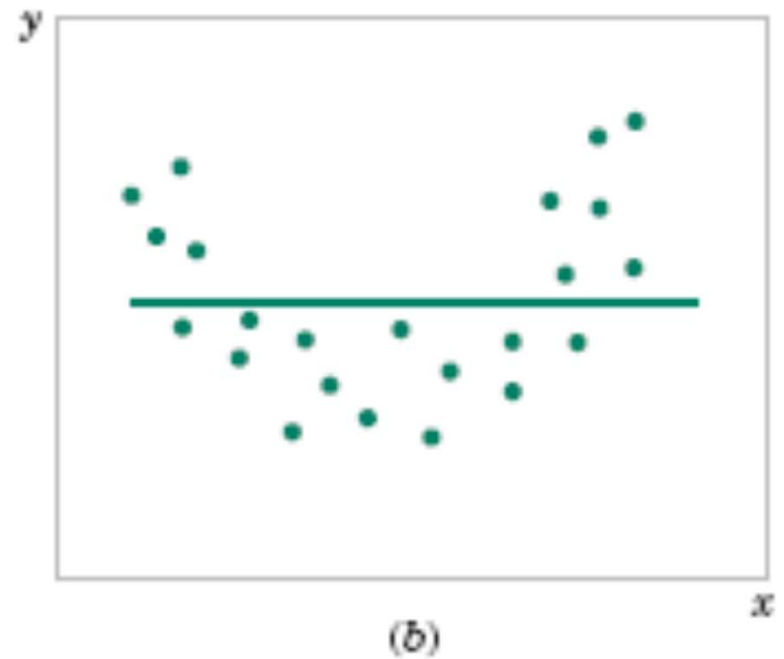
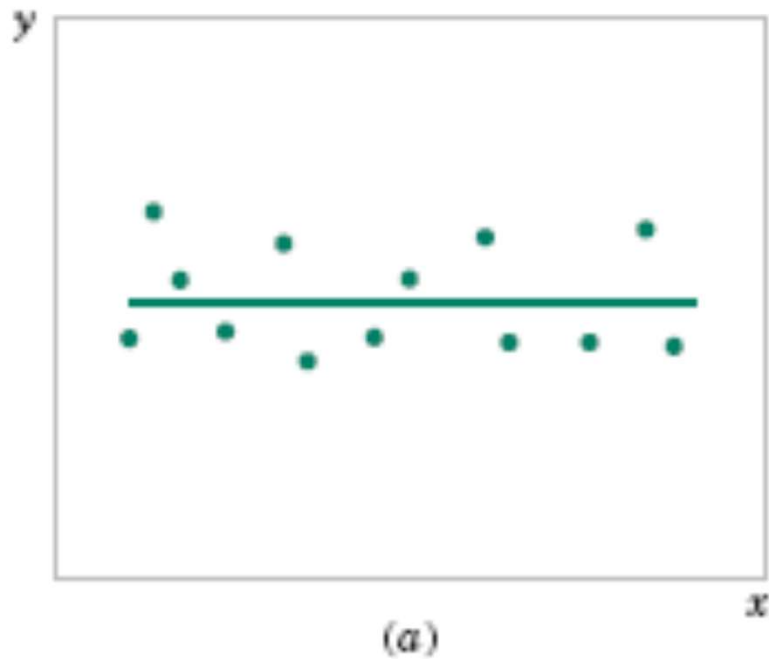
 Reject



$$\text{R.R.: } t \leq -t_{\alpha/2}$$

$$\text{or } t \geq t_{\alpha/2}$$

Hypothesis Tests in Simple Linear Regression



The hypothesis $H_0: \beta_1 = 0$ is not rejected.

Hypothesis Tests in Simple Linear Regression



(a)



(b)

The hypothesis $H_0: \beta_1 = 0$ is rejected.

Example: Oxygen Purity

Regression Analysis

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R-Sq (adj) = 87.1%

$$\frac{\hat{\beta}_1}{\sqrt{\hat{\sigma}^2/S_{xx}}} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)}$$

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ANalysis Of VAriance identity

$$E(SSE/(n-2)) = \sigma^2$$

$$SSE/\sigma^2 \sim \chi^2_{n-2}$$

$$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$$

$$SSR/\sigma^2 \sim \chi^2_1$$

If $\beta_1=0$, what is the distribution of $\frac{SSR/1}{SSE/(n-2)}$?

$$\frac{W/u}{Y/v} \sim F_{u,v} \quad W \sim \chi^2_u, \quad Y \sim \chi^2_v$$

ANalysis Of Variance (ANOVA) table for simple regression

<i>Source</i>	<i>df</i>	<i>Sum Sq.</i>	<i>Mean Sum Sq.</i>	<i>F</i>
<i>Regression</i>	<i>1</i>	<i>SSR</i>	$MSR = \frac{SSR}{1} = SSR$	$\frac{SSR}{MSE}$
<i>Error</i>	<i>n-2</i>	<i>SSE</i>	$MSE = \hat{\sigma}^2 = \frac{SSE}{n-2}$	
<i>Total</i>	<i>n-1</i>	<i>SST</i>		

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

can be tested via:

$$t^2 = f \Rightarrow t_{\alpha/2, n-2}^2 = F_{\alpha, 1, n-2}$$

reject H_0 if $F > F_{\alpha, 1, n-2}$.

equivalent to t test in simple linear regression. F-test is only two-sided.

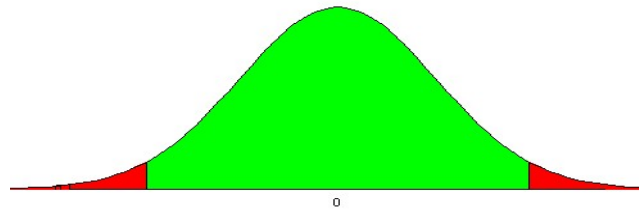
Is the linear regression model significant?

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

Test statistics:

$$T = \frac{b_1}{s_{b_1}} \sim t_{n-2}$$



Accept

Reject

$$\text{R.R. : } t \leq -t_{\alpha/2} \\ \text{or } t \geq t_{\alpha/2}$$

$$t^2 = f \Rightarrow t_{\alpha/2, n-2}^2 = F_{\alpha, 1, n-2}$$

reject H_0 if $F > F_{\alpha, 1, n-2}$.

only two-sided

Hypothesis testing on β_1

$$H_0: \beta_1 = 0$$

$$\text{Test statistics: } T = \frac{b_1}{s_{b_1}} \sim t_{n-2}$$

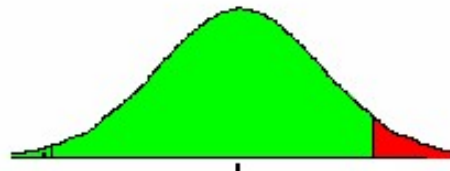
$$H_a: \beta_1 > 0$$

$$H_a: \beta_1 \neq 0$$

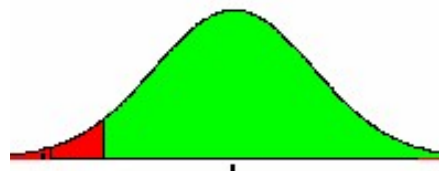
$$H_a: \beta_1 < 0$$

Accept

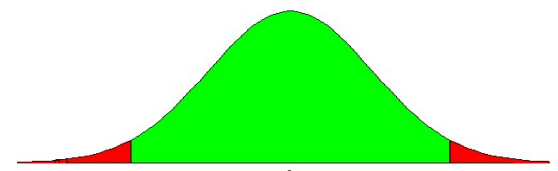
Reject



$$\text{R.R.: } t \geq t_\alpha$$



$$\text{R.R.: } t \leq -t_\alpha$$



$$\begin{aligned} \text{R.R.: } t &\leq -t_{\alpha/2} \\ \text{or } t &\geq t_{\alpha/2} \end{aligned}$$